

# Introduction to Survival Analysis

Pramesh Subedi, Ph.D.

Department of Mathematics  
Ohio University

Graduate Seminar – Fall 2026

- ➊ **Survival Data and Censoring**
- ➋ **Survival and Hazard Functions**
- ➌ **Nonparametric Estimation**
  - Kaplan–Meier estimator
  - Nelson–Aalen estimator
- ➍ **Comparing Survival Between Groups**
  - Log-rank test
- ➎ **Survival Regression**
  - Cox proportional hazards model
  - Additive hazards model

*Methods illustrated with applications in R*

# What Is Survival Analysis?

**Survival analysis** is a collection of statistical methods for analyzing data in which the outcome of interest is **time until an event occurs**.

The **survival time** is measured from a well-defined **time origin** to the occurrence of a specified **event of interest**.

## Examples:

- Birth  $\longrightarrow$  Death
- Birth  $\longrightarrow$  Cancer diagnosis
- Disease onset  $\longrightarrow$  Death
- Study entry  $\longrightarrow$  Relapse
- Marriage  $\longrightarrow$  Divorce

*Survival time is also referred to as failure time or time to event.*

A distinctive feature of survival data is that the event of interest may **not be observed for every individual**.

An observation is **censored** when the exact event time is unknown, but some information about the survival time is available.

## Common types of censoring:

- **Right censoring:** the event occurs after the last observed time
- **Left censoring:** the event occurred before the individual was first observed
- **Interval censoring:** the event occurred between two observation times

*Right censoring is the most common form in survival studies.*

# Right Censoring

A survival time is **right censored** when follow-up ends before the event of interest is observed.

$T^*$  = true survival time,       $C$  = censoring time

$$T = \min(T^*, C)$$

$$\delta = \begin{cases} 1, & \text{event observed,} \\ 0, & \text{censored.} \end{cases}$$

**Event Observed**

$$T^* \leq C$$

$$T = T^*, \quad \delta = 1$$

**Right Censored**

$$T^* > C$$

$$T = C, \quad \delta = 0$$

*Common reasons: study ends, withdrawal, or loss to follow-up.*

# Example: Leukemia Remission Times

**Treatment group:** 21 leukemia patients

Outcome: **time to relapse** (weeks)

**Relapse** | 6 6 6 7 10 13 16 22 23

**Censored** | 6<sup>+</sup> 9<sup>+</sup> 10<sup>+</sup> 11<sup>+</sup> 17<sup>+</sup> 19<sup>+</sup> 20<sup>+</sup> 25<sup>+</sup> 32<sup>+</sup> 32<sup>+</sup> 34<sup>+</sup> 35<sup>+</sup>

10

Relapse observed at 10 weeks

10<sup>+</sup>

Censored at 10 weeks

Observed data for each patient:  $(T, \delta)$

# Why Survival Methods?

Survival data contain both

## Observed events

$$T = T^*$$

Exact event time is known.

## Censored observations

$$T = C, \quad T^* > C$$

Exact event time is unknown.

Censored observations are **not missing observations**: they tell us that the individual remained event-free for at least the observed follow-up time.

Survival methods use information from **both events and censored observations**.

# The Survival Function

Let  $T^*$  denote the true survival time.

The **survival function** is

$$S(t) = P(T^* > t)$$

the probability of remaining **event-free beyond time**  $t$ .

For example,

$$S(5) = 0.80$$

means that the probability of remaining event-free beyond 5 years is **80%**.

**Properties:**

$$0 \leq S(t) \leq 1, \quad S(0) = 1, \quad S(t) \text{ is non-increasing.}$$

# The Hazard Rate

The survival function describes the probability of remaining **event-free beyond time  $t$** . In contrast, the **hazard rate** describes the instantaneous rate of experiencing the event at time  $t$ , among individuals who have remained event-free up to that time.

For a continuous survival time  $T^*$ ,

$$\alpha(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T^* < t + \Delta t \mid T^* \geq t)}{\Delta t}$$

Using the survival function,

$$\alpha(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \frac{S(t) - S(t + \Delta t)}{S(t)} = -\frac{S'(t)}{S(t)} = -\frac{d}{dt} \log S(t).$$

The **cumulative hazard function** is

$$A(t) = \int_0^t \alpha(s) ds.$$

# Connecting the Hazard and Survival Functions

$$\alpha(t) = -\frac{S'(t)}{S(t)} = -\frac{d}{dt} \log S(t)$$

$$\Rightarrow A(t) = \int_0^t \alpha(s) ds = -\int_0^t \frac{d}{ds} \log S(s) ds$$

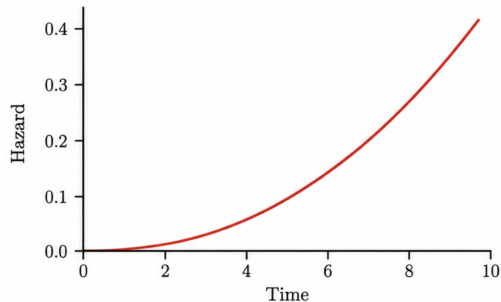
$$\Rightarrow A(t) = -\log S(t) \quad (S(0) = 1)$$

$$S(t) = e^{-A(t)} = \exp \left\{ -\int_0^t \alpha(s) ds \right\}$$

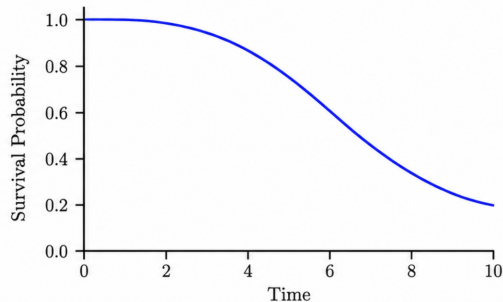
$$\alpha(t) \longrightarrow A(t) \longleftrightarrow S(t)$$

# Hazard and Survival: A Visual View

**Hazard Function  $\alpha(t)$**



**Survival Function  $S(t)$**



## Key Connection

As the hazard accumulates over time, the survival probability decreases:

$$S(t) = \exp \left\{ - \int_0^t \alpha(s) ds \right\}$$

# The Risk Set

At any time  $t$ , the **risk set** consists of individuals who

- have **not yet experienced the event**, and
- have **not yet been censored**.

Let

$$Y(t) = \text{number of individuals at risk just before time } t$$

As events and censoring occur,

Initial cohort  $\longrightarrow$  smaller risk sets over time.

At each event time  $t_j$ , survival methods use

$$Y(t_j) = \text{number at risk}, \quad d_j = \text{number of events.}$$

# The Kaplan–Meier Estimator

The survival function

$$S(t) = P(T^* > t)$$

is unknown and must be estimated from the observed survival data.

Suppose the distinct observed event times are

$$t_1 < t_2 < \cdots < t_k.$$

At each event time  $t_j$ , let

$$Y(t_j) = \text{number at risk just before } t_j, \quad d_j = \text{number of events at } t_j.$$

Then the conditional probability of surviving past  $t_j$  is

$$1 - \frac{d_j}{Y(t_j)}.$$

The **Kaplan–Meier estimator** is

$$\hat{S}(t) = \prod_{t_j \leq t} \left( 1 - \frac{d_j}{Y(t_j)} \right)$$

# Kaplan–Meier Example: Leukemia Remission Data

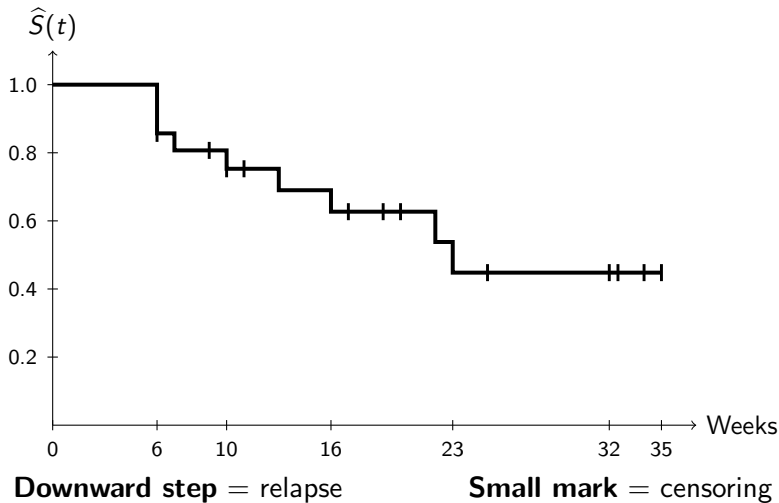
6, 6, 6, 7, 10, 13, 16, 22, 23, 6<sup>+</sup>, 9<sup>+</sup>, 10<sup>+</sup>, 11<sup>+</sup>, 17<sup>+</sup>, 19<sup>+</sup>, 20<sup>+</sup>, 25<sup>+</sup>, 32<sup>+</sup>, 32<sup>+</sup>, 34<sup>+</sup>, 35<sup>+</sup>

| $t_j$ | $Y(t_j)$ | $d_j$ | $c_j$ | $1 - \frac{d_j}{Y(t_j)}$ | $\hat{S}(t_j)$                                                                                               |
|-------|----------|-------|-------|--------------------------|--------------------------------------------------------------------------------------------------------------|
| 0     | 21       | 0     | 0     | 1                        | 1                                                                                                            |
| 6     | 21       | 3     | 1     | $\frac{18}{21}$          | $\frac{18}{21} = 0.8571$                                                                                     |
| 7     | 17       | 1     | 1     | $\frac{16}{17}$          | $\frac{18}{21} \times \frac{16}{17} = 0.8067$                                                                |
| 10    | 15       | 1     | 2     | $\frac{14}{15}$          | $\frac{18}{21} \times \frac{16}{17} \times \frac{14}{15} = 0.7529$                                           |
| 13    | 12       | 1     | 0     | $\frac{11}{12}$          | $\frac{18}{21} \times \frac{16}{17} \times \frac{14}{15} \times \frac{11}{12} = 0.6901$                      |
| 16    | 11       | 1     | 3     | $\frac{10}{11}$          | $\frac{18}{21} \times \frac{16}{17} \times \frac{14}{15} \times \frac{11}{12} \times \frac{10}{11} = 0.6274$ |

$c_j$  = number censored after  $t_j$  and before the next relapse time.

# The Kaplan–Meier Survival Curve

For the leukemia treatment group, the Kaplan–Meier estimate is a **step function** that changes only at observed relapse times.



# The Nelson–Aalen Estimator

Recall that the **cumulative hazard function** is

$$A(t) = \int_0^t \alpha(s) ds.$$

At each observed event time  $t_j$ , let

$$Y(t_j) = \text{number at risk just before } t_j, \quad d_j = \text{number of events at } t_j.$$

The **Nelson–Aalen estimator** of the cumulative hazard is

$$\hat{A}(t) = \sum_{t_j \leq t} \frac{d_j}{Y(t_j)}$$

Thus, at each event time, the cumulative hazard increases by

$$\Delta \hat{A}(t_j) = \frac{d_j}{Y(t_j)}$$

$\hat{A}(t)$  is a non-decreasing step function.

# Nelson–Aalen Example: Leukemia Remission Data

Using the same leukemia treatment group:

6, 6, 6, 7, 10, 13, 16, 22, 23, 6<sup>+</sup>, 9<sup>+</sup>, 10<sup>+</sup>, 11<sup>+</sup>, 17<sup>+</sup>, 19<sup>+</sup>, 20<sup>+</sup>, 25<sup>+</sup>, 32<sup>+</sup>, 32<sup>+</sup>, 34<sup>+</sup>, 35<sup>+</sup>

| $t_j$ | $Y(t_j)$ | $d_j$ | $c_j$ | $\frac{d_j}{Y(t_j)}$ | $\hat{A}(t_j)$                         |
|-------|----------|-------|-------|----------------------|----------------------------------------|
| 0     | 21       | 0     | 0     | 0                    | 0                                      |
| 6     | 21       | 3     | 1     | $\frac{3}{21}$       | $\frac{3}{21} = 0.1429$                |
| 7     | 17       | 1     | 1     | $\frac{1}{17}$       | $\frac{3}{21} + \frac{1}{17} = 0.2017$ |
| 10    | 15       | 1     | 2     | $\frac{1}{15}$       | $0.2017 + \frac{1}{15} = 0.2683$       |
| 13    | 12       | 1     | 0     | $\frac{1}{12}$       | $0.2683 + \frac{1}{12} = 0.3517$       |
| 16    | 11       | 1     | 3     | $\frac{1}{11}$       | $0.3517 + \frac{1}{11} = 0.4426$       |

$c_j$  = number censored after  $t_j$  and before the next relapse time.

# Nelson–Aalen Cumulative Hazard Curve

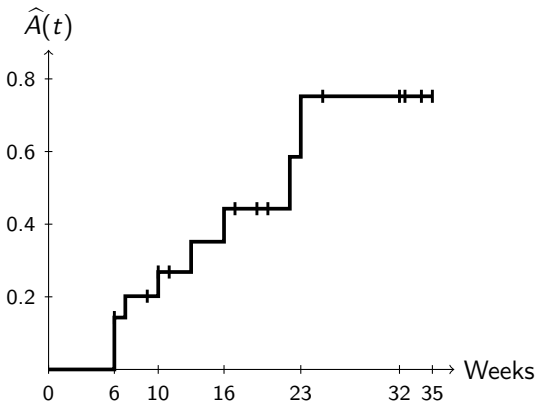
For the leukemia treatment group, the estimated cumulative hazard is a **step function** that increases only at observed relapse times.

## Cumulative hazard estimates

| $t$ | $\hat{A}(t)$ |
|-----|--------------|
| 0   | 0.000        |
| 6   | 0.143        |
| 7   | 0.202        |
| 10  | 0.268        |
| 13  | 0.352        |
| 16  | 0.443        |
| 22  | 0.585        |
| 23  | 0.752        |

- Increases at **relapse times**.
- Does not increase at censoring times.

**Reading the Curve** Each upward step represents an observed **relapse**. Censoring does not increase  $\hat{A}(t)$ , but reduces the subsequent risk set.



# Comparing Survival Between Groups

Kaplan–Meier curves can be estimated separately for different groups.

For example,

$$\hat{S}_1(t) \quad \text{and} \quad \hat{S}_2(t)$$

may represent survival under two different treatments.

This allows us to examine whether the groups have different **survival experiences over time**.

|                                         |   |                    |
|-----------------------------------------|---|--------------------|
| Estimate group-specific survival curves | → | Compare the curves |
|-----------------------------------------|---|--------------------|

A formal comparison can be made using the **log-rank test**.

# The Log-Rank Test

The **log-rank test** compares the survival experience of two or more groups over the entire follow-up period.

$$H_0 : S_1(t) = S_2(t) = \cdots = S_g(t) \quad \text{for all } t$$

$$H_A : \text{at least one survival function differs.}$$

At each observed event time, the test compares

|                           |     |                                       |
|---------------------------|-----|---------------------------------------|
| Observed number of events | vs. | Expected number of events under $H_0$ |
|---------------------------|-----|---------------------------------------|

The contributions across event times are combined into a test statistic that, under  $H_0$ , is approximately

$$\chi^2_{g-1}.$$

# Kaplan–Meier and Nelson–Aalen in R

**R package:** survival

```
library(survival)
```

## Basic syntax

```
Surv(time, status)
```

```
survfit(Surv(time, status) ~ 1, data = dataset)
```

- time: observed survival or follow-up time
- status: event indicator (1 = event, 0 = censored)
- data: specifies the data set

## Kaplan–Meier

```
km_fit <- survfit(  
  Surv(time, status) ~ 1,  
  data = dataset  
)
```

## Nelson–Aalen

```
na_fit <- survfit(  
  Surv(time, status) ~ 1,  
  data = dataset,  
  type = "fleming-harrington"  
)
```

# Kaplan–Meier Application: Lung Cancer Data

**R package:** survival

**Data:** lung

$n = 228$ , deaths = 165

**R command:**

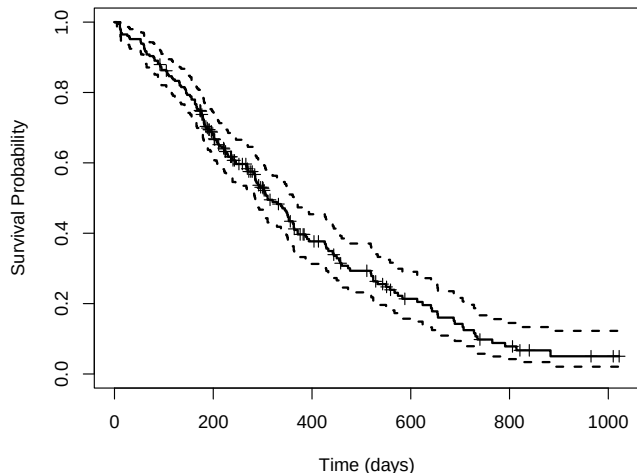
```
survfit(Surv(time,status) ~ 1)
```

**Estimated survival**

| Time     | $\hat{S}(t)$ |
|----------|--------------|
| 180 days | 0.722        |
| 365 days | 0.409        |
| 730 days | 0.116        |

**Median survival** 310 days

95% CI : (285, 363)



# Comparing Survival Between Groups in R

The formula in `survfit()` determines whether we estimate one overall survival curve or separate curves by group.

## One overall survival curve

```
survfit(Surv(time, status) ~ 1,  
        data = dat  
)
```

## Formal comparison: Log-rank test

```
survdif(Surv(time, status) ~ group,  
        data = dat  
)
```

`survfit()` **describes** the survival experience, while `survdif()` formally **compares** survival between groups.

## Separate curves by group

```
survfit( Surv(time, status) ~ group,  
        data = dat  
)
```

# Nelson–Aalen Application: Lung Cancer Data

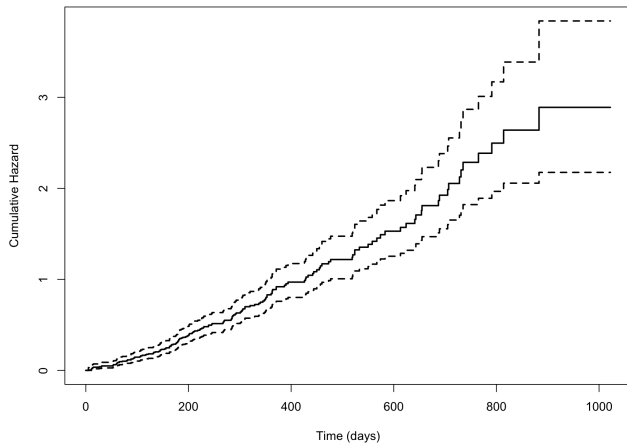
## R code

```
na_fit <- survfit(  
  Surv(time, status) ~ 1,  
  data = lung,  
  type = "fleming-harrington"  
)
```

```
plot(na_fit, fun = "cumhaz")
```

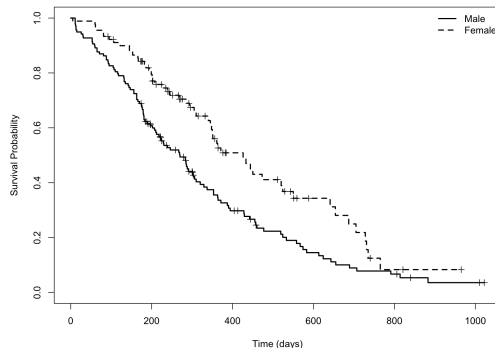
## Estimated cumulative hazard

| Time     | $\hat{A}(t)$ |
|----------|--------------|
| 180 days | 0.325        |
| 365 days | 0.888        |
| 730 days | 2.125        |



$\hat{A}(t)$  increases as observed events accumulate over time.

# Kaplan–Meier Survival Curves by Sex



| Group  | <i>n</i> | Deaths | Median Survival (95% CI) |
|--------|----------|--------|--------------------------|
| Male   | 138      | 112    | 270 days (212–310)       |
| Female | 90       | 53     | 426 days (348–550)       |

Females show **higher estimated survival** than males over much of the follow-up period.<sub>25 / 34</sub>

# Comparing Survival: The Log-Rank Test

The **log-rank test** compares the survival experience of two or more groups over the entire follow-up period.

$$H_0 : S_{\text{Male}}(t) = S_{\text{Female}}(t) \quad \text{for all } t$$

The test compares the **observed** and **expected** numbers of events under  $H_0$ :

| Group  | Observed Deaths | Expected Deaths |
|--------|-----------------|-----------------|
| Male   | 112             | 91.6            |
| Female | 53              | 73.4            |

For the lung cancer data,

$$\chi^2 = 10.3, \quad df = 1, \quad p = 0.001$$

There is **strong evidence that survival differs by sex**; the Kaplan–Meier curves indicate **better survival among females**.

# The Cox Proportional Hazards Model

Kaplan–Meier and Nelson–Aalen describe survival without incorporating covariates.

The **Cox proportional hazards model** relates the hazard to a set of covariates:

$$\alpha(t | \mathbf{x}) = \alpha_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x})$$

The model separates the hazard into

$$\underbrace{\alpha_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp(\boldsymbol{\beta}^T \mathbf{x})}_{\text{covariate effects}} .$$

where

- $\alpha_0(t)$  is the **baseline hazard**,
- $\mathbf{x}$  is the vector of covariates,
- $\boldsymbol{\beta}$  is the vector of regression coefficients.

The baseline hazard  $\alpha_0(t)$  is left **unspecified**, so the Cox model is **semiparametric**.

# Hazard Ratios in the Cox Model

For two individuals with covariate values  $\mathbf{x}_1$  and  $\mathbf{x}_2$ ,

$$\frac{\alpha(t | \mathbf{x}_1)}{\alpha(t | \mathbf{x}_2)} = \frac{\alpha_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x}_1)}{\alpha_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x}_2)}$$

$$HR = \exp \left\{ \boldsymbol{\beta}^T (\mathbf{x}_1 - \mathbf{x}_2) \right\}$$

For a one-unit increase in covariate  $x_j$ ,

$$HR = e^{\beta_j}$$

- $HR > 1$ : higher hazard of the event,
- $HR < 1$ : lower hazard of the event,
- $HR = 1$ : no difference in hazard.

The baseline hazard  $\alpha_0(t)$  **cancels**, so covariate effects are interpreted through **hazard ratios**.

# The Proportional Hazards Assumption

For two individuals with covariate values  $\mathbf{x}_1$  and  $\mathbf{x}_2$ ,

$$\frac{\alpha(t \mid \mathbf{x}_1)}{\alpha(t \mid \mathbf{x}_2)} = \exp \left\{ \boldsymbol{\beta}^T (\mathbf{x}_1 - \mathbf{x}_2) \right\}.$$

The right-hand side does **not depend on time**.

Therefore,

The hazard ratio between two individuals is constant over time.

**Proportional hazards does not mean constant hazards.**

The baseline hazard  $\alpha_0(t)$  may change freely over time, but the hazards for different covariate patterns remain proportional:

$$\alpha(t \mid \mathbf{x}) = \alpha_0(t) \times \text{time-constant relative effect}$$

# Fitting the Cox Model in R

Using the lung cancer data, we model survival as a function of **sex and age**:

```
cox_fit <- coxph(  
  Surv(time, status) ~ sex_group + age,  
  data = lung  
)
```

| Covariate       | HR    | 95% CI         | p-value |
|-----------------|-------|----------------|---------|
| Female vs. Male | 0.599 | (0.431, 0.831) | 0.002   |
| Age (per year)  | 1.017 | (0.999, 1.036) | 0.065   |

For sex,

$$HR = e^{-0.513} = 0.599.$$

After adjusting for age, females have approximately a **40% lower hazard of death** than males.

# Checking the Proportional Hazards Assumption

The proportional hazards assumption can be assessed using **Schoenfeld residuals**.

In R:

```
ph_test <- cox.zph(cox_fit)
ph_test
```

| Covariate | $\chi^2$ | df | p-value |
|-----------|----------|----|---------|
| Sex       | 2.608    | 1  | 0.11    |
| Age       | 0.209    | 1  | 0.65    |
| Global    | 2.771    | 2  | 0.25    |

$H_0$  : proportional hazards assumption holds

There is **no evidence of a violation** of the proportional hazards assumption (**global**  $p = 0.25$ ).

# The Additive Hazards Model

The Cox model describes covariate effects **multiplicatively** on the hazard:

$$\alpha(t | \mathbf{x}) = \alpha_0(t) \exp(\boldsymbol{\beta}^T \mathbf{x}).$$

An alternative is the **additive hazards model**:

$$\alpha(t | \mathbf{x}) = \alpha_0(t) + \boldsymbol{\beta}^T \mathbf{x}$$

For a one-unit increase in  $x_j$ , holding other covariates fixed,

$$\alpha(t | x_j + 1) - \alpha(t | x_j) = \beta_j$$

Thus,  $\beta_j$  represents an **absolute difference in hazard**, rather than a hazard ratio.

|                  | Effect Scale   | Interpretation             |
|------------------|----------------|----------------------------|
| Cox model        | Multiplicative | Relative hazard            |
| Additive hazards | Additive       | Absolute hazard difference |

# Fitting the Additive Hazards Model in R

The additive hazards model can be fitted using `aalen()` from the `timereg` package.  
`library(timereg)`

```
aalen_fit <- aalen(  
  Surv(time, status) ~ const(sex_group) + const(age),  
  data = lung,  
  robust = TRUE  
)
```

```
summary(aalen_fit)
```

| Covariate       | $\hat{\beta}$ | 95% CI                | p-value |
|-----------------|---------------|-----------------------|---------|
| Female vs. Male | -0.00115      | (-0.00185, -0.00045)  | 0.001   |
| Age (per year)  | 0.000038      | (-0.000003, 0.000078) | 0.069   |

After adjusting for age, females have an estimated **absolute hazard of death 0.00115 per person-day lower** than males.

# Thank You

Questions and Discussion

Pramesh Subedi, Ph.D.

Department of Mathematics  
Ohio University